

## SEEDLINE · PROBLEM STATEMENT

# SeedLine problem statement

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## DOCUMENT CONTROL

|             |                                       |         |                |
|-------------|---------------------------------------|---------|----------------|
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| Project TRL | <b>TRL 3</b> Proof of concept         |         |                |
| Repository  | github.com/BoujeeEnjinia1701/seedline |         |                |
| Source      | docs/01-problem.md @ uncommitted      |         |                |

## REVISION HISTORY

| Version | Date       | Author       | Change                                                                                          |
|---------|------------|--------------|-------------------------------------------------------------------------------------------------|
| 0.3     | 2026-09-25 | Amish Chadha | Record the first user group and row-spacing kit decisions (SDL-DDR-001); update open questions  |
| 0.2     | 2026-09-25 | Amish Chadha | Populate to TRL 2 (problem, users, context, constraints, out of scope, prior work with sources) |
| 0.1     | 2026-09-24 | Amish Chadha | Initial scaffold                                                                                |

## SeedLine problem statement

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Hand-seeding is slow and uneven, and precision planters are too costly for small plots. The gap is a push seeder that places one seed at a time at a set spacing, costs about the same as the cheapest plate seeder, and can be adapted to any crop by printing a new metering plate instead of buying one.

### ■ The problem

Most of the world's farms are small. About 84 % of the world's roughly 570 million farms are under 2 ha ([Our World in Data](#), from [Lowder et al.](#)). On these plots and on market gardens, seed is still sown by hand with a hoe, a stick or a jab planter, or dribbled from a cheap plate seeder.

Three problems follow:

1. **Labor.** Hand planting with a hoe has been estimated at about 56 h per hectare for one worker in southern Africa, and a jab planter cuts that to about a third ([Baudron et al., cited in a CSBE jab planter study](#)). Planting happens in a short window after the first rains, so slow planting delays part of the crop.
2. **Uneven spacing.** Hand-dropped seed lands in clumps and gaps. Uneven stands cost yield: Purdue trials on maize found a loss of about 2.2 to 2.5 bu/acre (about 140 to 160 kg/ha) for each inch (25 mm) increase in the standard deviation of plant-to-plant spacing ([Nielsen, Purdue University](#)). Clumped vegetable seed also has to be thinned by hand.
3. **Cost and fit of existing tools.** Low-cost plate seeders such as the Earthway 1001-B (about \$187 with six plates) pick up several seeds at a time, so crops usually need thinning, while singulating roller seeders such as the Jang JP-1 cost about \$499 before rollers at about \$25 each ([University of Minnesota Extension, 2024](#)). Users also report that the Earthway is inaccurate with small seed and has no in-row spacing adjustment ([Paperpot Co.](#)). A farmer whose crop or seed size does not match a stock plate has no easy way to get one.

SeedLine is a single-row push seeder whose ground wheel drives a vertical metering plate through a chain. The plate is 3D printed for each crop and seed size, so a farmer, a cooperative or a local maker with a printer can make a plate for any seed. An optional row-marker kit sets the spacing between rows.

**First users (decided by Amish, 2026-09-25, SDL-DDR-001):** smallholder field crops, meaning maize, beans, sorghum and groundnut on 0.75 m rows. Market-garden vegetables follow as a second plate set, with seed under 2 mm sown pelleted. The row-spacing kit is a marker arm.

## ■ Users and context

| User                                      | Need                                                                                                                 | Context                                                                           |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Smallholder farmer                        | Plant maize, beans, sorghum or groundnuts at the right spacing in the planting window, with less labor and less seed | Plots of 0.2 to 2 ha, hand-tilled or ox-ploughed soil, rain-fed, often one worker |
| Market gardener                           | Sow vegetables in beds without thinning, at low cost                                                                 | Beds of 0.75 to 1.2 m, fine tilled soil, 3 to 6 rows per bed                      |
| Cooperative or agro-dealer with a printer | Print plates on demand for members' crops and varieties; lend or rent seeders                                        | Rural town, one FDM printer, basic tools                                          |
| Local welder or maker                     | Build and repair the frame from common steel tube and bicycle parts                                                  | Workshop with a welder or drill, bicycle spares available                         |
| Extension worker or researcher            | Recommend spacings and check stands                                                                                  | Field demonstrations, trial plots                                                 |

## Operating environment

- **Soil:** tilled loam to sandy loam, with clods, crop residue and stones up to about 50 mm; seedbed quality varies widely on hand-tilled plots.
- **Seed:** from maize and beans (8 to 15 mm) down to vegetable seed of 1 to 2 mm; some seed is treated with fungicide or insecticide.
- **Climate:** 10 to 40 °C, strong sun, dust; rain and mud at planting time.
- **Supply chain:** steel tube, bicycle chain and sprockets, bearings and wheelbarrow or trolley wheels are common in regional towns; 3D printers are uncommon in villages but increasingly present in towns, universities and maker spaces.

## ■ Constraints

- Garage-buildable prototype, about \$400 USD ( `project.yaml` ).
- A replicated base seeder (without the marker kit) should cost no more than a low-cost plate seeder, about \$200 in parts.
- Metering plates printable in PLA, PETG or ASA on a common desktop FDM printer, without supports.
- Frame and drive from common steel sections, bicycle chain parts and bearings that local workshops stock.
- Pushed by one person at walking speed on a tilled seedbed.
- No engine, battery or electronics in the base design.

## ■ Out of scope

- Tractor or animal-drawn planters.
- Fertilizer placement (a later option, not part of this concept).
- Transplanting, seed-tape or paper-pot systems.
- Seed treatment, seed supply or agronomic advice beyond the plate spacing tables.
- Vacuum or air-assisted metering.

## ■ Prior work

- **Commercial push seeders.** The Earthway plate seeder and the Jang roller seeder set the price and performance range for push seeders ([UMN Extension](#); [Paperpot Co.](#)). Both use interchangeable plates or rollers bought from the maker.
- **Open 3D-printed seeders.** The Hour Farm published a hackable, 3D-printed precision seeder with interchangeable printed seed discs on FarmHack in 2019 under CC BY 4.0 ([FarmHack](#)). It shows that printed discs work for vegetable seed at bed scale.
- **Printed parts in precision metering.** A Norfolk farmer printed singulator discs and sprockets for a six-row vacuum maize planter, reporting about 99 % singulation and no visible wear after 70 ha ([3D Printing Industry, 2019](#)). This is a tractor planter, but it supports the durability of printed metering parts.
- **Jab planters.** Hand jab planters are promoted for conservation agriculture in Africa and cut planting time to about a third of hoe planting ([CSBE study](#); [FAO jab planter user manual](#)). They place seed hill by hill, so spacing depends on the operator.
- **Research push planters.** Many university studies build and test manual single-row planters with plate or cell-wheel metering (for example, [a manually operated planter for different seeds](#)). Their field capacity and seed placement data have not yet been reviewed against SDL-CAL-001; that review remains open.

## ■ Open questions

- Which partner and region for co-design and field trials? Proposed, awaiting Amish; under the portfolio rule of 2026-09-25, community designs pick co-design partners per area later.
- Who prints plates in practice: the farmer, a cooperative, an agro-dealer or a central maker who mails them?
- How common is treated seed among target users, and what handling rules apply?
- Do market gardeners want two or three metering units ganged on one frame (the later bed-seeding variant)?

## ■ User research and co-design

Requirements must come from the people who will use the seeder.

- [ ] Identify a local partner organization (farmer cooperative, extension service, NGO or university)
- [ ] Run co-design sessions with intended users; record who, where and what was learned
- [ ] Validate crop list, spacing, soil and cost assumptions in the field
- [ ] Revise requirements (REQ) from findings before freezing the design